
Modular Robotics and Locomotion: Application to Limbless Robots

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Ph.D. Thesis



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Outline

1. **Introduction and Objectives**
2. Classification
3. Locomotion in 1D
4. Locomotion in 2D
5. Minimal configurations
6. Experiments
7. Conclusions and future work

The Locomotion Problem

- Development of a very versatile robot with the full capability of moving on different terrains.

- **Higher Level:**

- Perception of the environment
- Planning trajectories
- Navigation
- Making decisions

- **Lower Level:**

- Coordination of the articulations
- Robot morphology (Limbs, wheels,...)
- Gaits

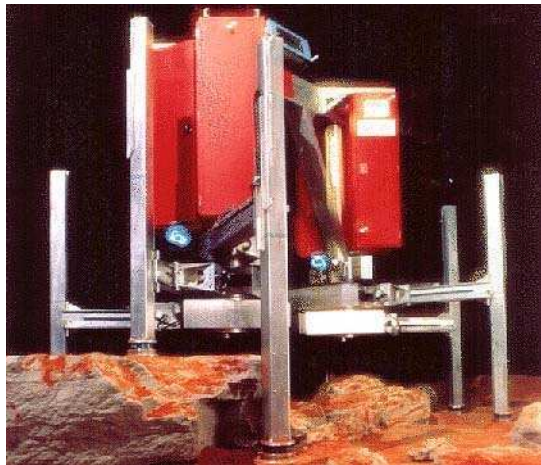


Scope of this thesis

The locomotion problem (II)

Classic approach:

- Study the terrain
- Design the robot's structure
- Gait realization

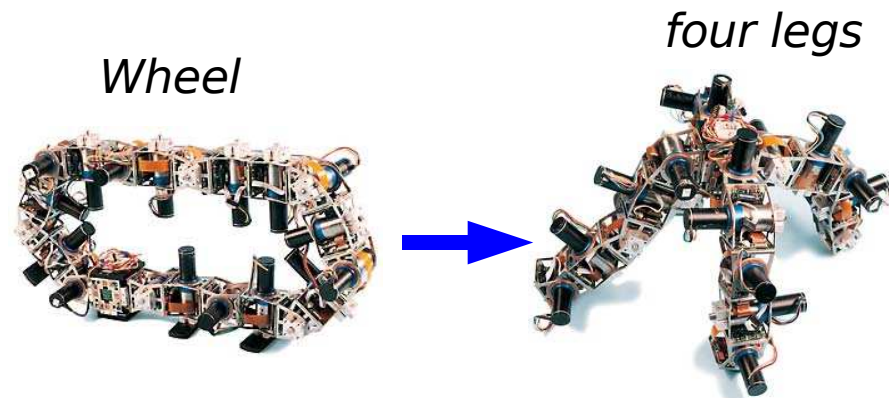


(CMU Ambler, Krotkov et al, 1995)

New approach: (Yim, 1995) Self-reconfigurable Modular Robots

- Consist of equal modules
- Their shape is adapted to the terrain

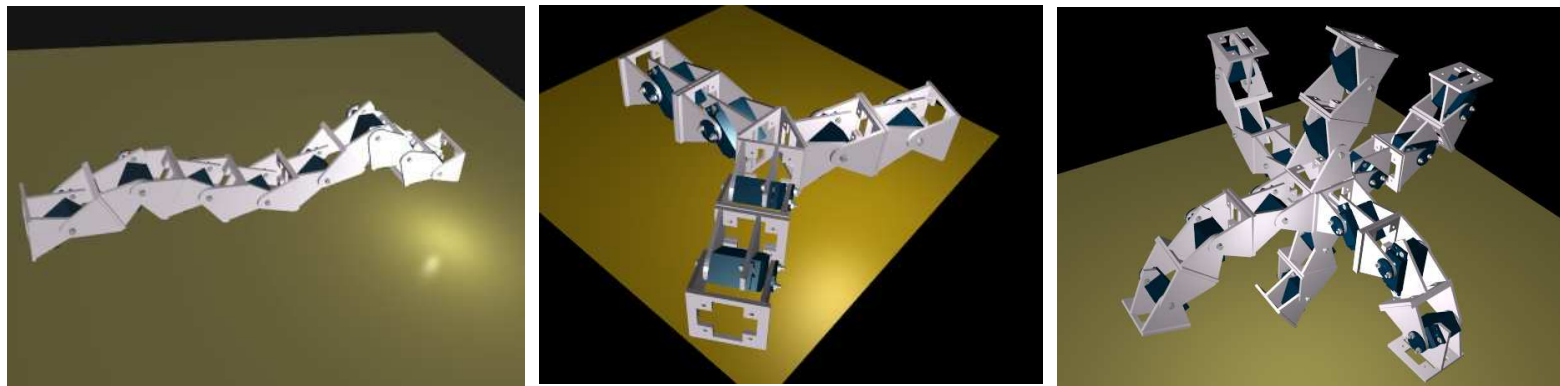
Polybot experiment



(Polybot, Yim et al, 2000)

Morphology

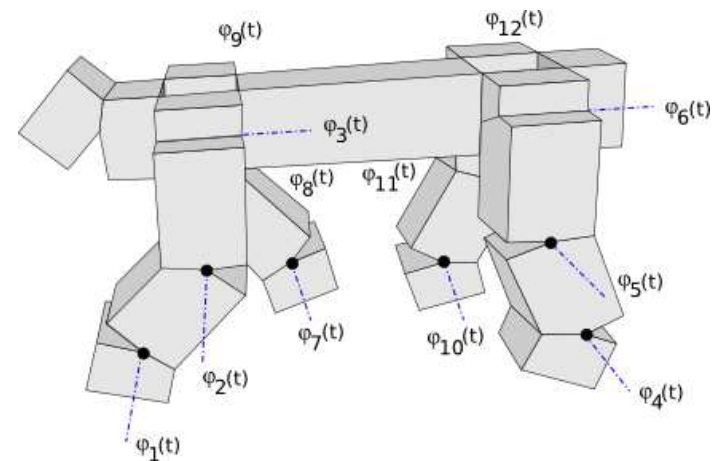
- Each morphology has its own locomotion capabilities that should be studied
- The number of configurations growth exponentially with the number of modules
- Solution: A Classification should be established
- This dissertation is focused on the **1D topology modular robot group**



Controller

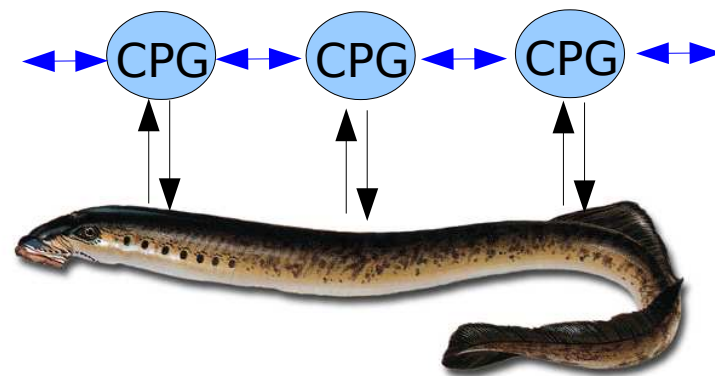
- **Coordination problem:**

Calculation of the joint's angles to realize a gait: $\varphi_i(t)$



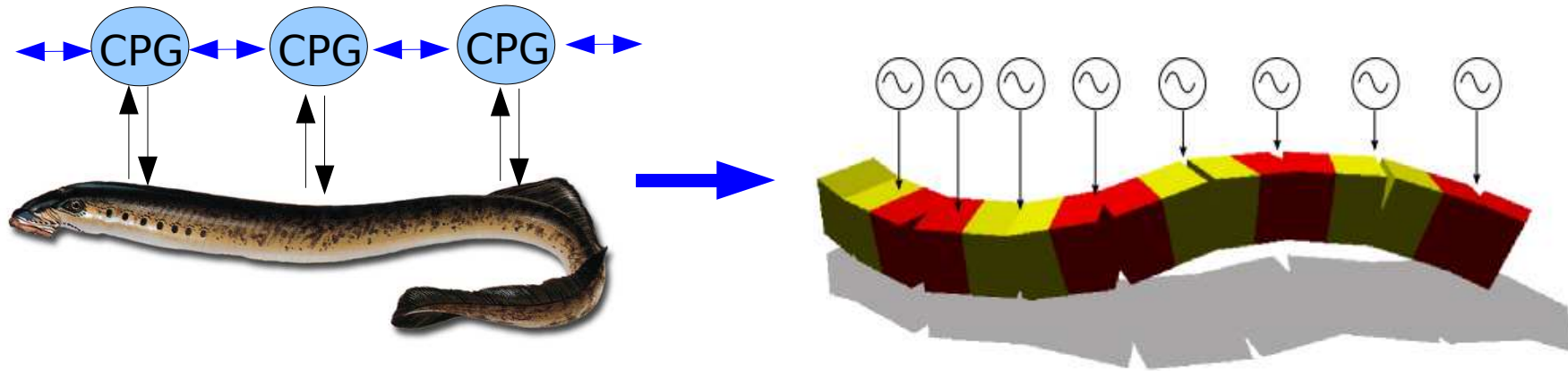
- **Classic approach:** Mathematical modelling
 - Calculation by inverse kinematics
 - Disadvantages: The equations are only valid for an specific morphology

- **Bio-inspired controllers:** CPGs
 - Central Pattern Generators
 - CPGs control the rhythmic activities
 - Ej. The locomotion of the lamprey

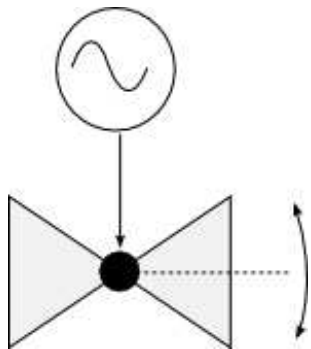


Hypothesis: Sinusoidal oscillators

- CPGs are replaced by a **Simplified model**



- Sinusoidal oscillators:

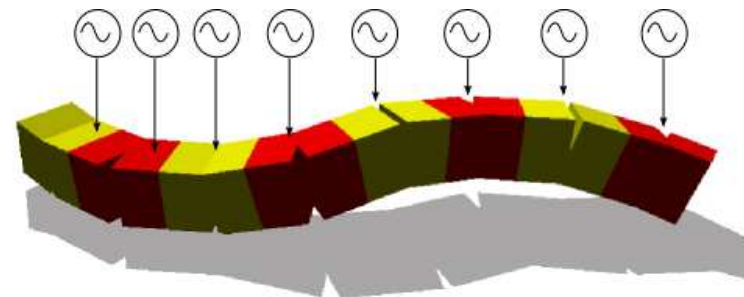


$$\varphi_i(t) = A_i \sin\left(\frac{2\pi}{T}t + \psi_i\right) + O_i$$

- Advantages:**
 - Few resources required

Objectives

- Feasibility of the control model
- Locomotion gaits
- Characterization
- Minimal configurations
- Kinematics

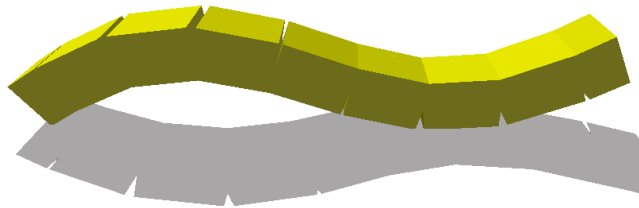


Study the locomotion of the 1D Topology pitch-pitch and pitch-yaw connecting modular robots of any length in one and two dimensions

Restrictions

- Groups of study:

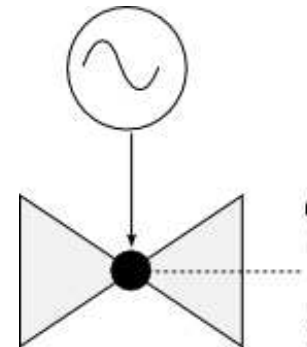
Pitch-pitch connection



Pitch-yaw connection



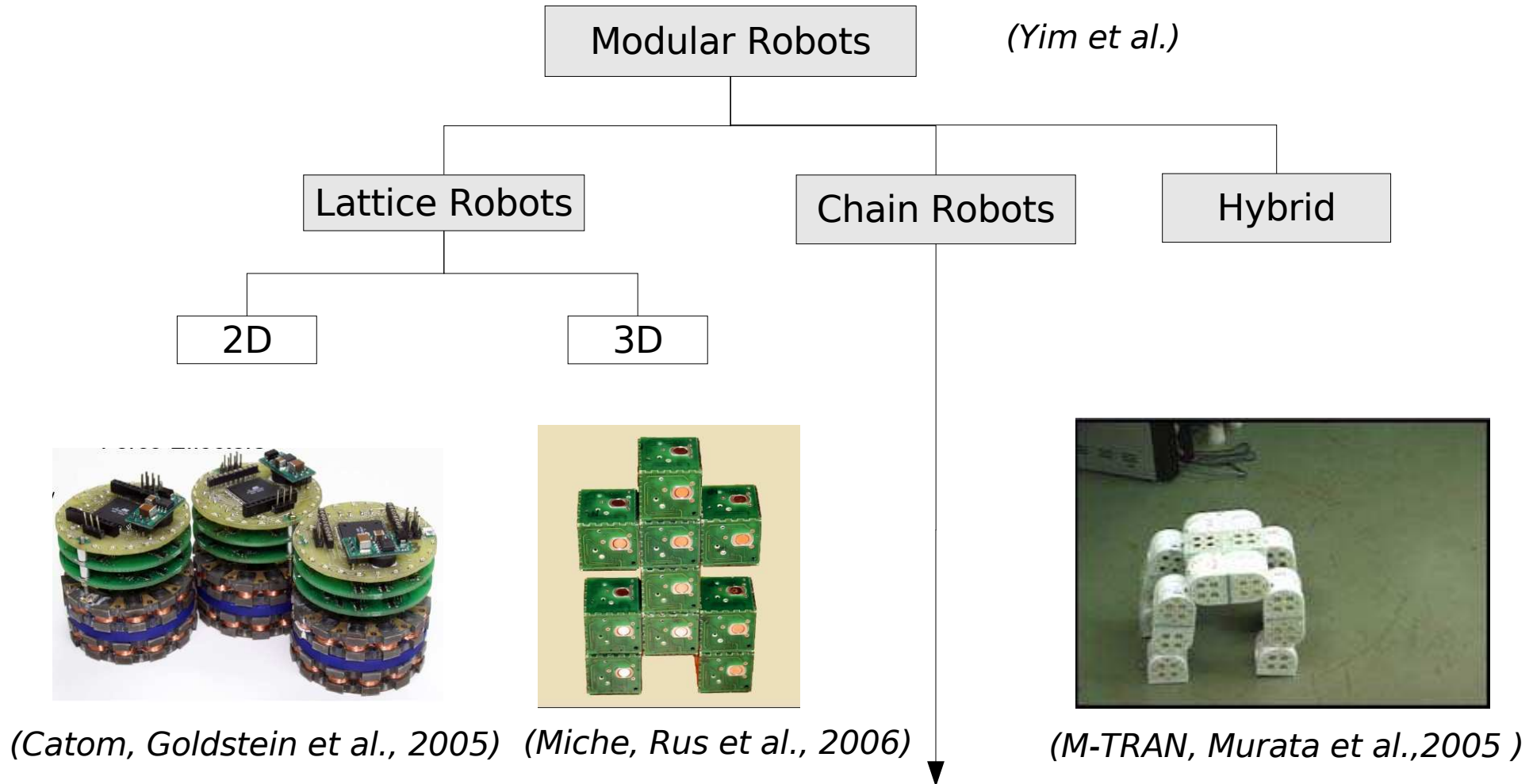
- Steady state
- Flat homogeneous surfaces without obstacles
- Open loop control
- Modules without sensors



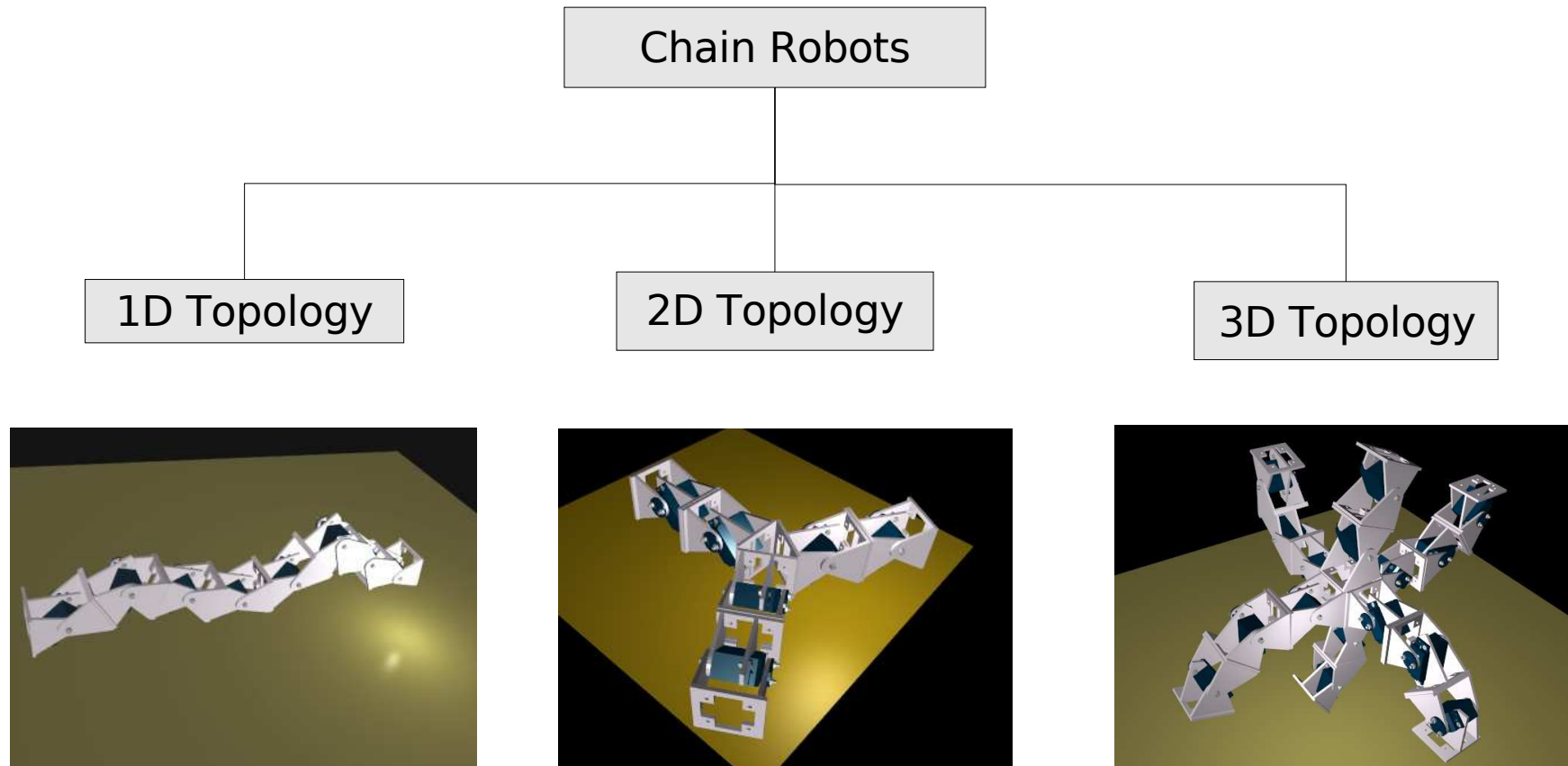
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1. Introduction and objectives
2. **Classification**
3. Locomotion in 1D
4. Locomotion in 2D
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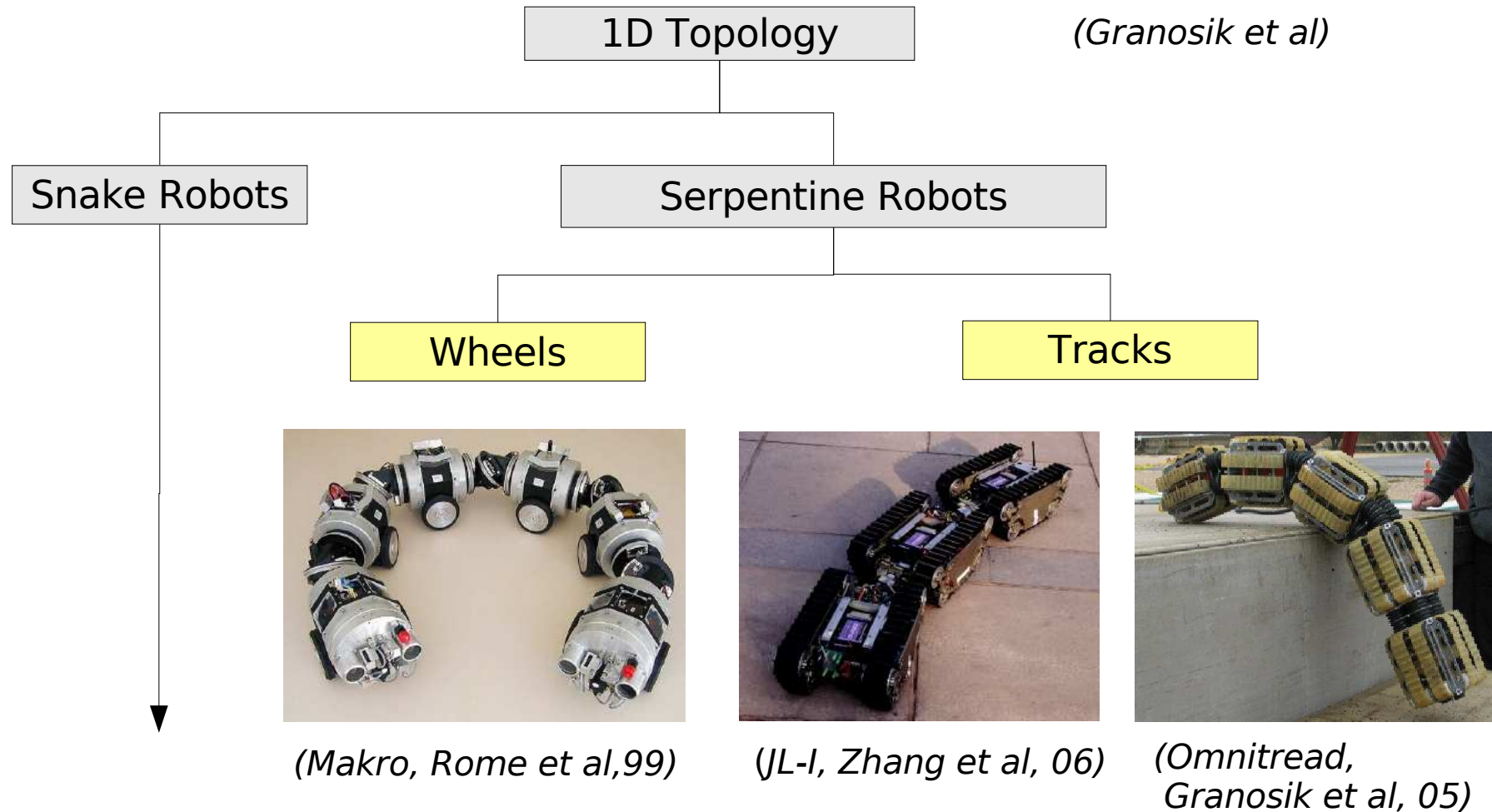
Modular Robots classification (I)



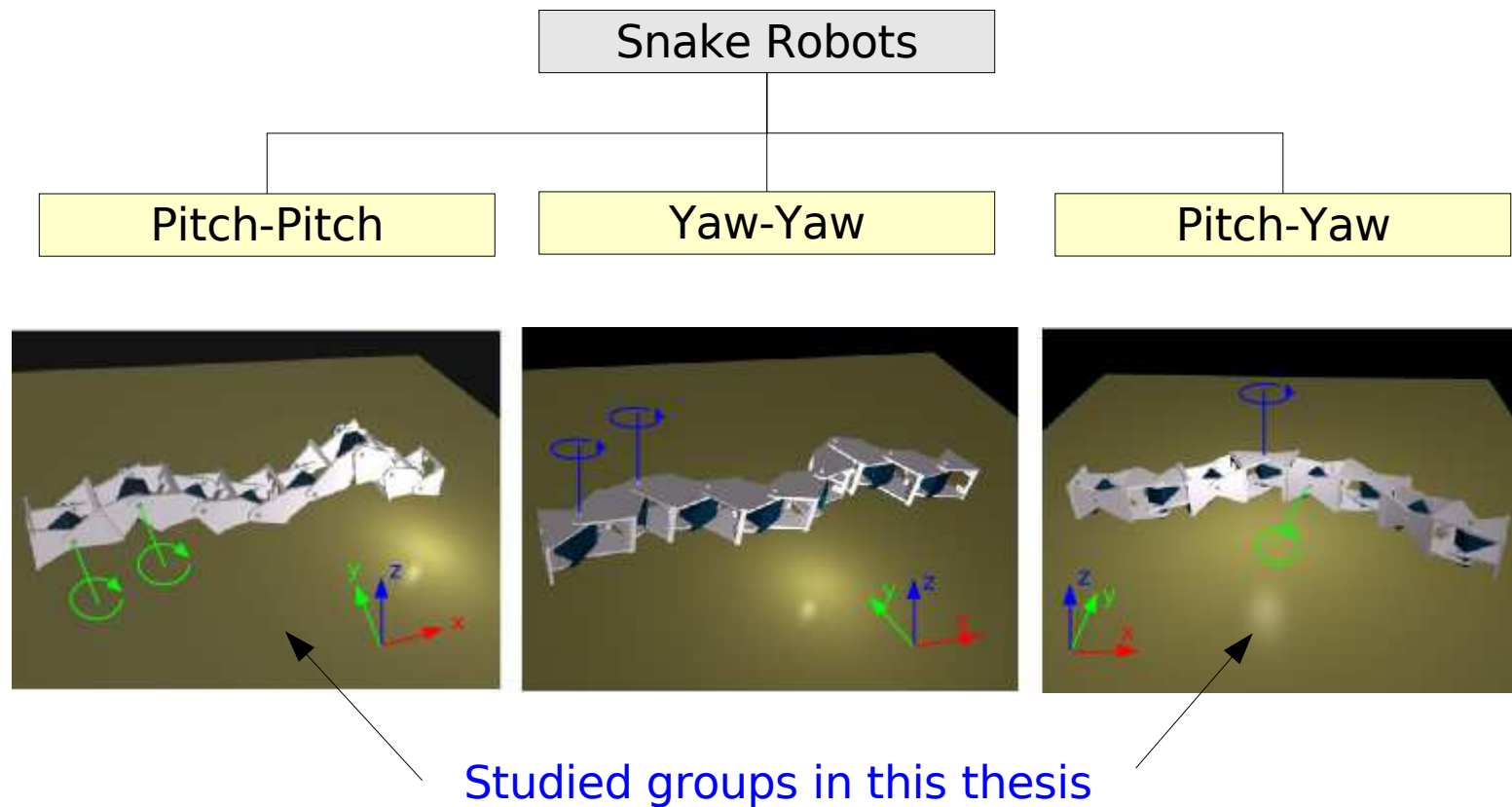
Modular Robots classification (II)



Modular Robots classification (III)



Modular Robots classification (IV)

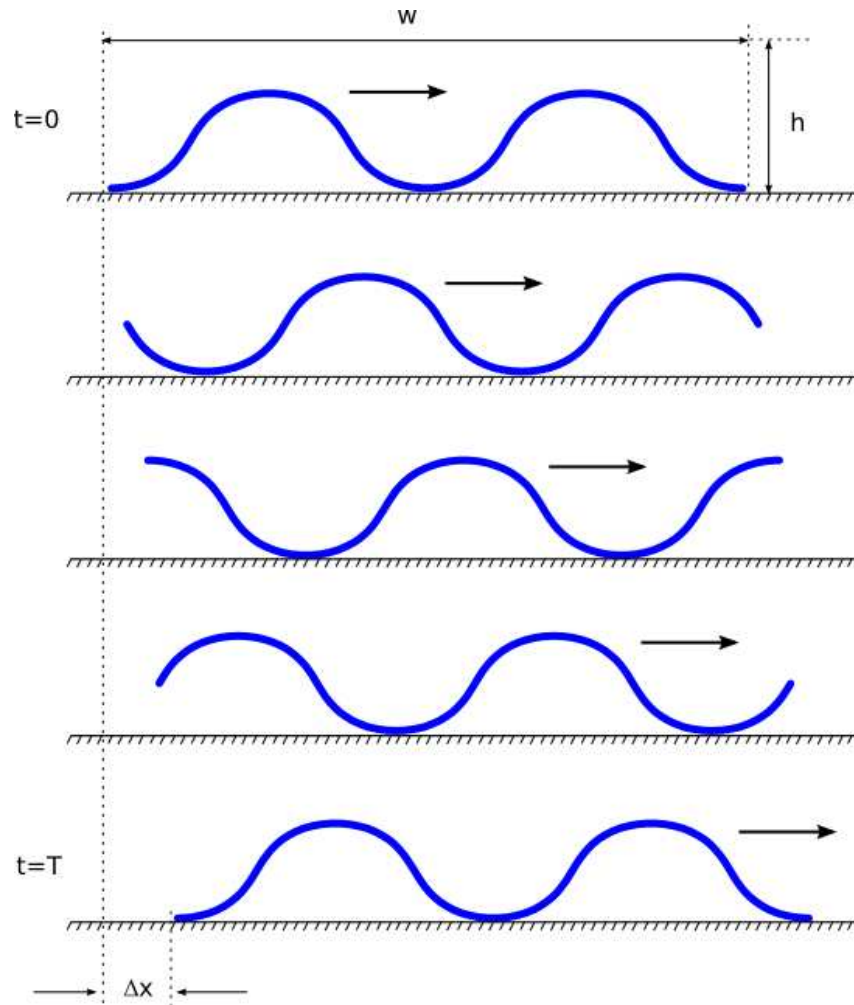


(The whole classification map can be found in page 46)

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Locomotion mechanism

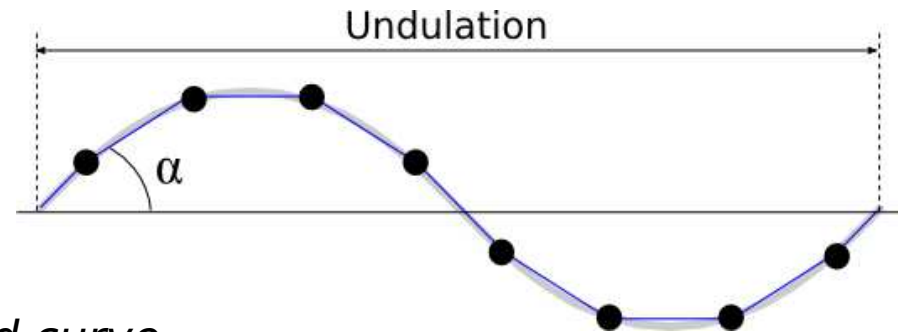


- Locomotion performed by the body wave propagation
- Step: Δx
- Mean Speed: $V = \frac{\Delta x}{T}$
- Size: width (w) x height (h)

Shape of the Body wave

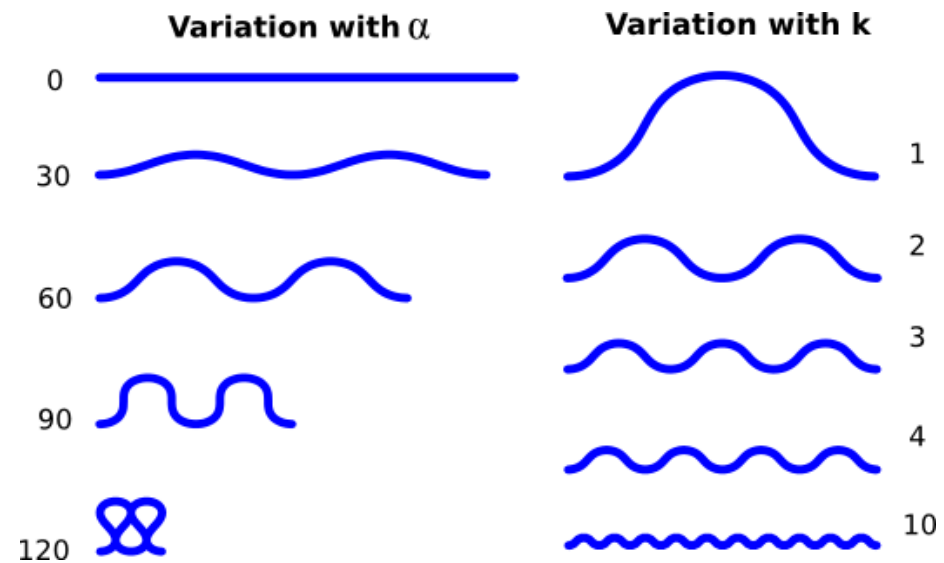
- **Serpenoid curve**

- *When Sin. oscillations applied*
- *(Hirose, 1975)*
- *Snakes: Horizontal serpenoid curve*
- *Pitch-pitch group: Vertical serpenoid curve*



- **Parameters:**

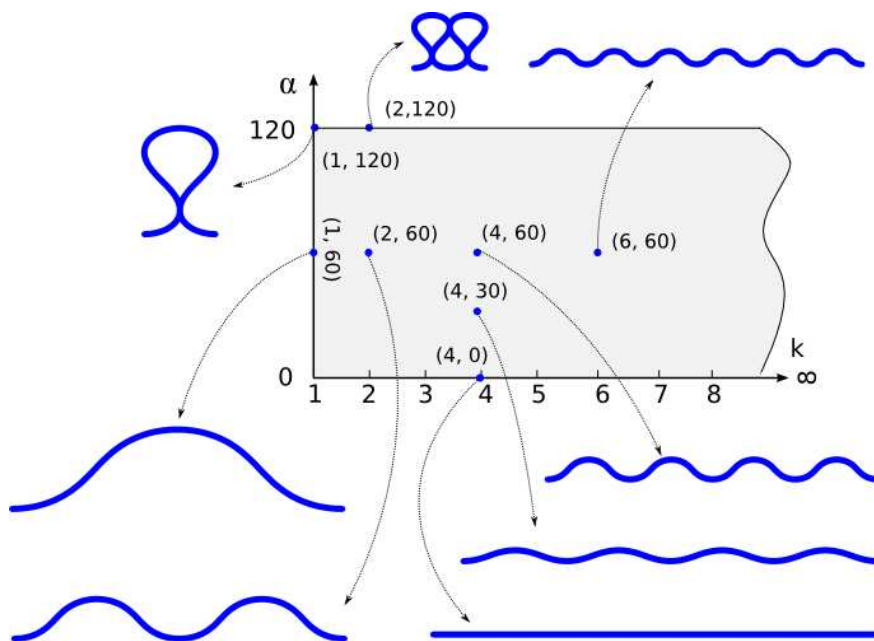
- Winding angle: α
- Number of undulations: k



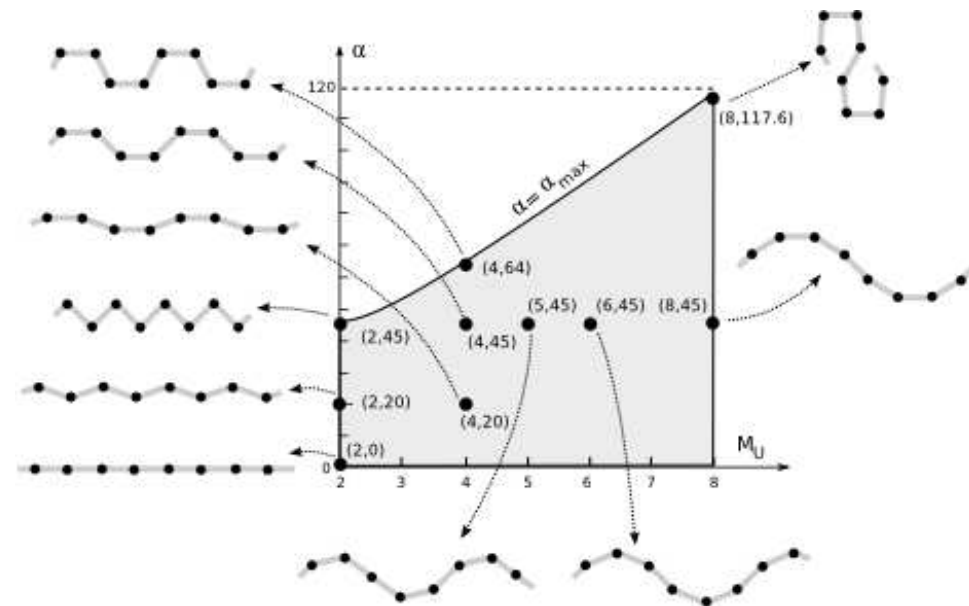
Shape space

- We propose to represent the all the body waves as points in the **shape space**

Continuous



Discrete

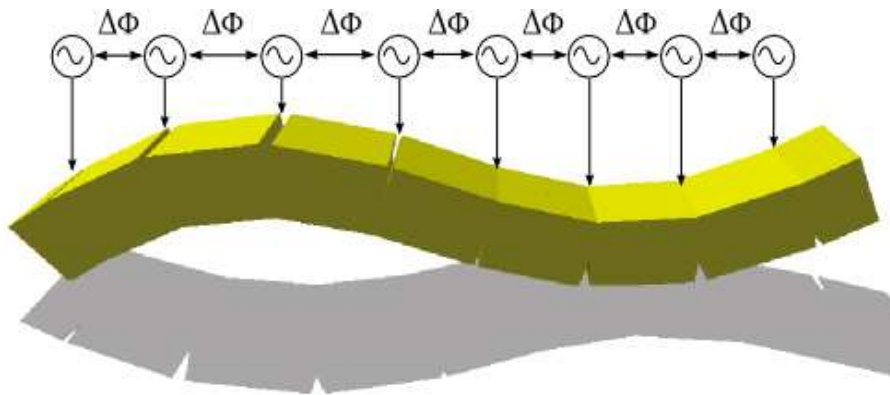


Ej. $M=8$

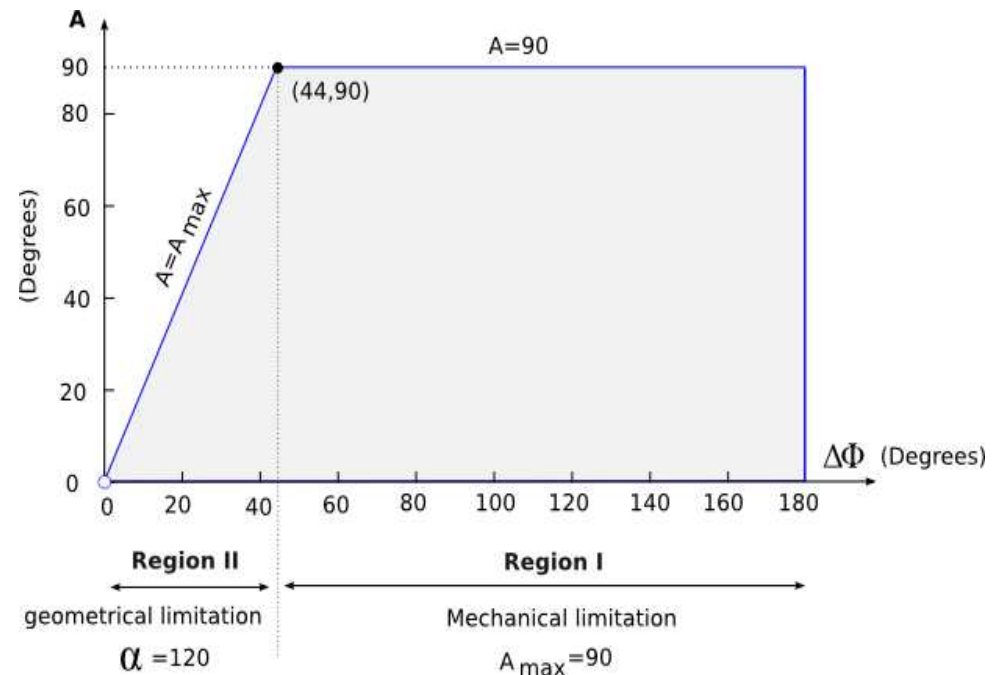
$$M_u = \frac{M}{k}$$

Control space

- Robot are controlled by means of M equal sinusoidal oscillators
- The same frequency
- The same amplitude A
- The same phase difference $\Delta\phi$

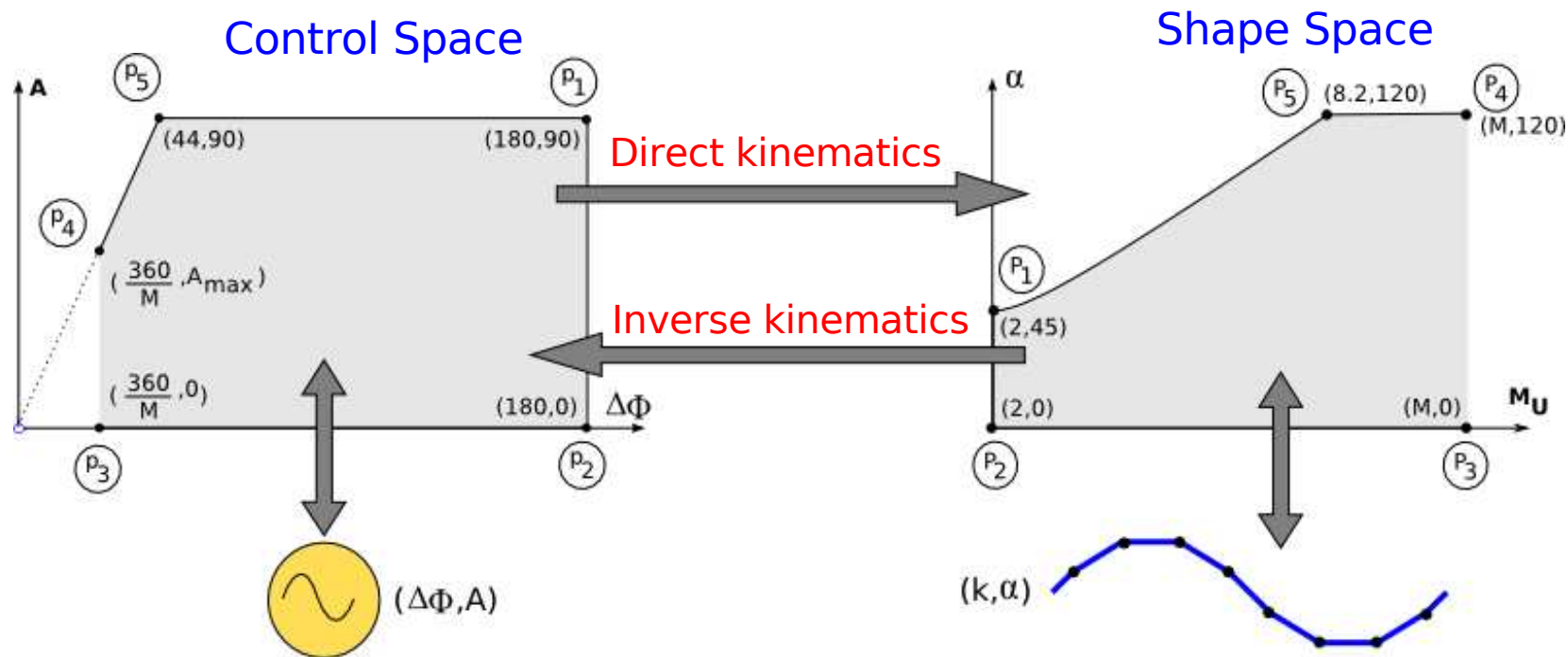


- Only two parameters are needed:
 - Amplitude: A
 - Phase difference: $\Delta\phi$



Kinematics

- Direct and Inverse Kinematics problems
- Solutions by means of a space transformation
- The constraints are set in the shape space (robot dimensions and step)



Kinematics (II)

- **Transformations:**

$$A = 2\alpha \sin\left(\frac{\pi k}{M}\right) \quad \Delta\Phi = \frac{360 k}{M}$$

- **Dimensions:**

$$w = k \int_0^{\frac{l}{k}} \cos\left(\alpha \cos\left(\frac{2\pi k}{l} s\right)\right) ds$$

$$h = \int_0^{\frac{l}{4k}} \sin\left(\alpha \cos\left(\frac{2\pi k}{l} s\right)\right) ds$$

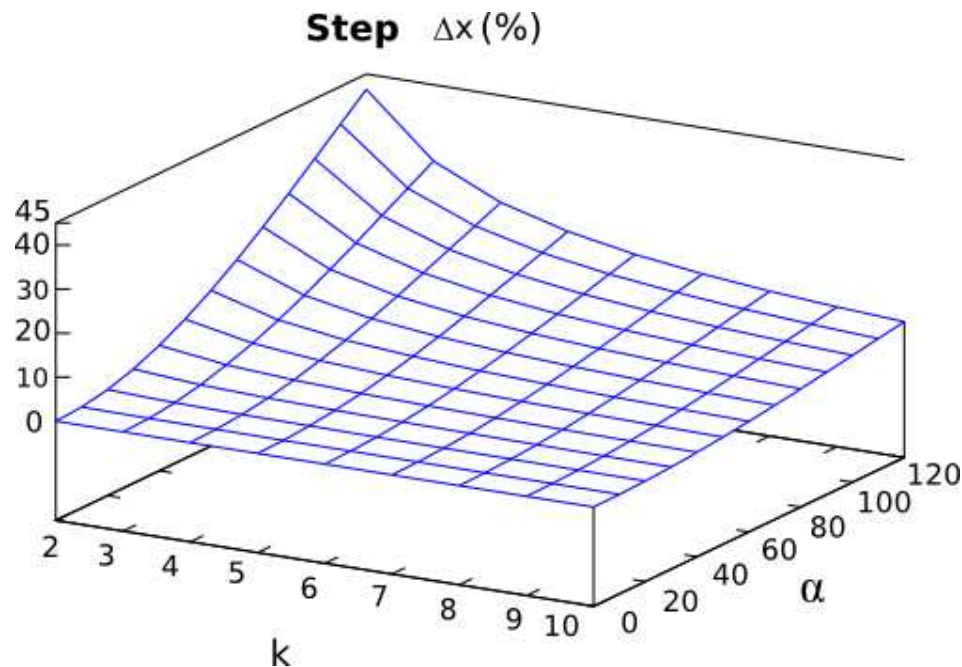
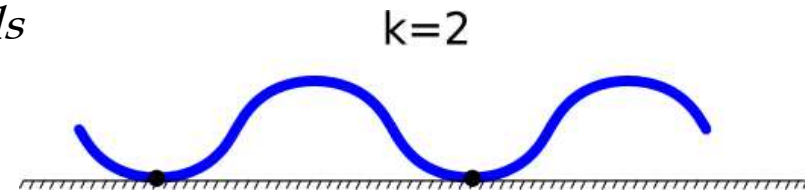
- **Step:**

$$\Delta x = \frac{l}{k} - \int_0^{\frac{l}{k}} \cos\left(\alpha \cos\left(\frac{2\pi k}{l} s\right)\right) ds$$

No analytical solutions

Step characterization

- Stability: $k \geq 2$
- Step equation: $\Delta x = \frac{l}{k} - \int_0^{\frac{k}{l}} \cos\left(\alpha \cos\left(\frac{2\pi k}{l} s\right)\right) ds$
- Deduction from the robot shape



- The step increases with α
- The step decreases with k

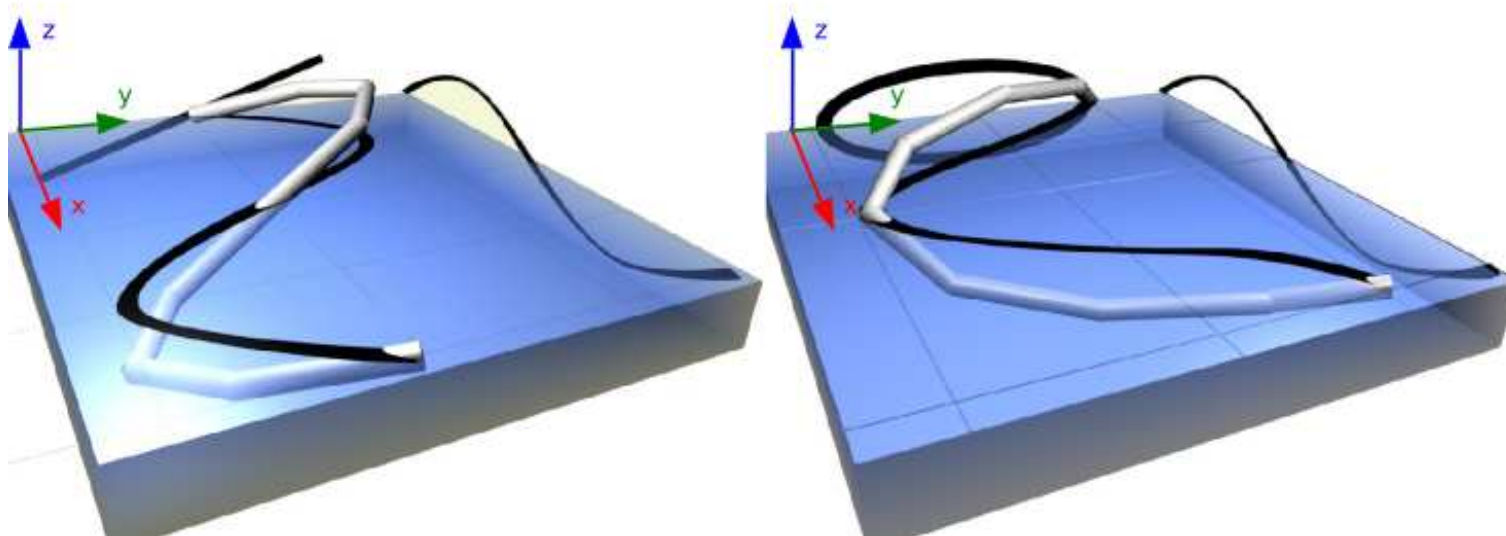
Biggest step:
 $\alpha = 120, k = 2$

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Locomotion mechanism

- 3D Body wave propagation
- Linear Step: Δr
- Angular Step: $\Delta \gamma$
- Dimensions: width (w) x length (l_x) x height (h)



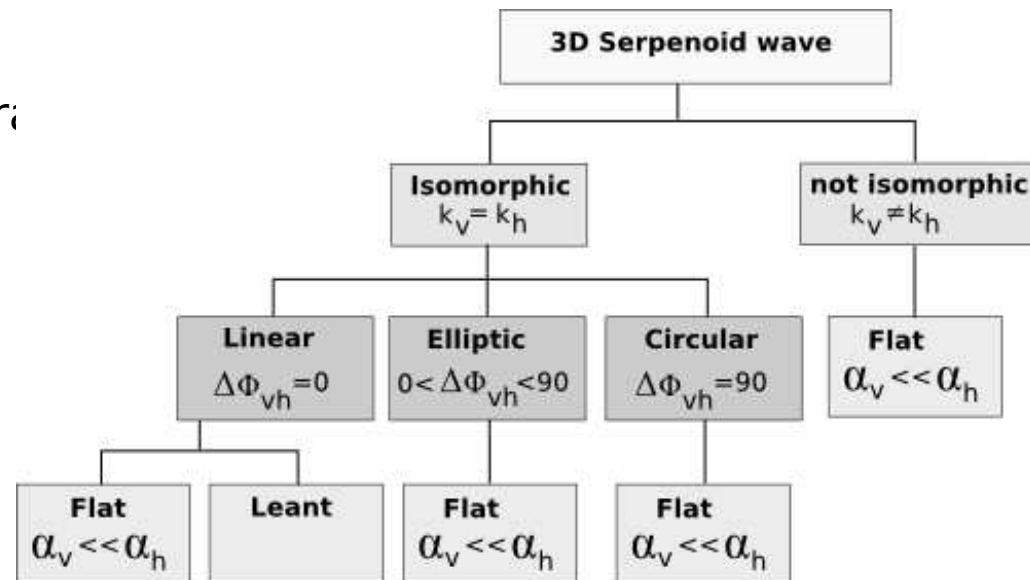
Shape space

- Superposition of two bidimensional waves:
 - Vertical wave: α_v, k_v
 - Horizontal wave: α_h, k_h
 - Phase difference: $\Delta\Phi_{vh}$
- The relationship between the parameters determines the type of wave

The Shape space has
5 dimensions

$$\alpha_v, \alpha_h, k_v, k_h, \Delta\Phi_{vh}$$

TYPE OF WAVES

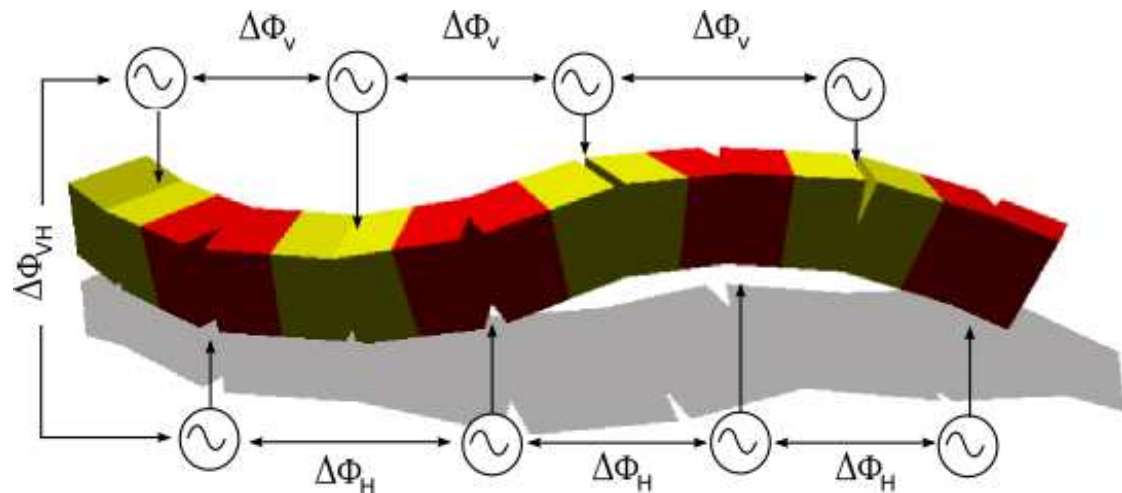


Control space

- Vertical and horizontal oscillators
- Equal horizontal oscillators: $A_h, \Delta\Phi_h$
- Equal vertical oscillators: $A_v, \Delta\Phi_v$
- Phase difference between vertical and horizontal: $\Delta\Phi_{vh}$
- The same period T

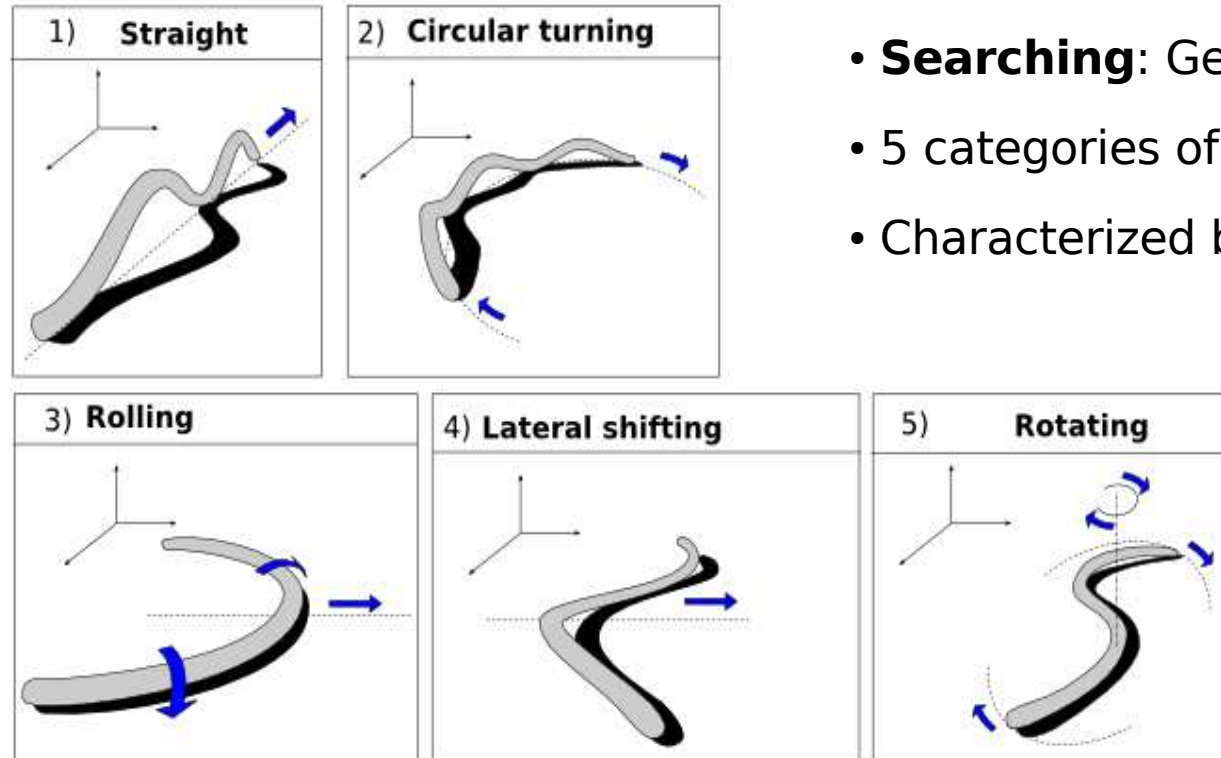
The control space
has **5 dimensions**

$$A_h, A_v, \Delta\Phi_h, \Delta\Phi_v, \Delta\Phi_{vh}$$



Locomotion gaits

Categories of gaits

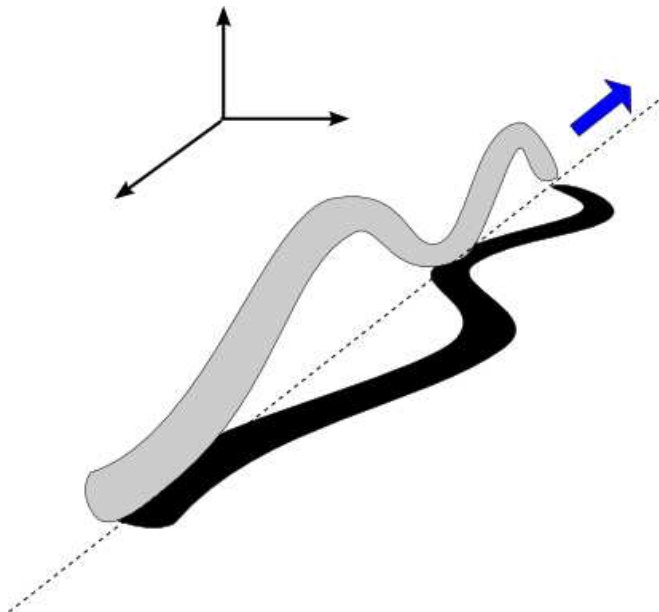


- **Searching:** Genetic algorithms
- 5 categories of gaits
- Characterized by the 3D body wave

Locomotion gaits (II)

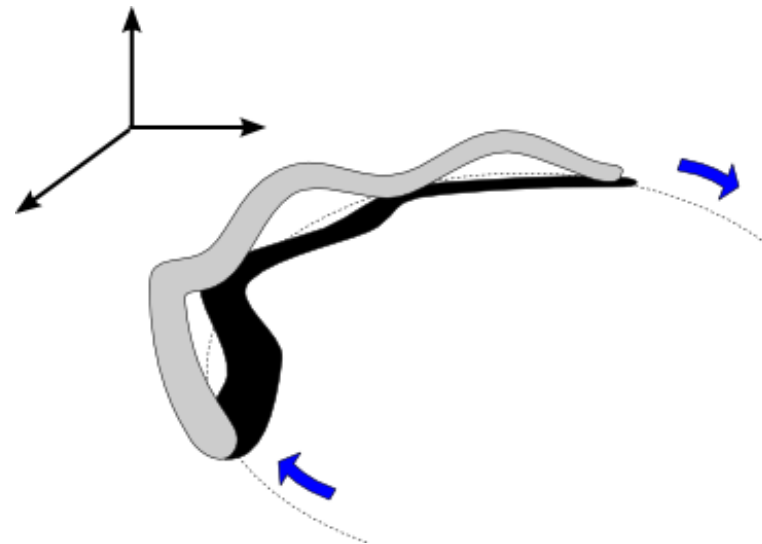
Straight

- Horizontal modules: $\varphi_h = 0$
- Parameters: α_v, k_v



Circular turning

- Horizontal modules: $\varphi_h \neq 0$
- Parameters: α_v, k_v, α_h



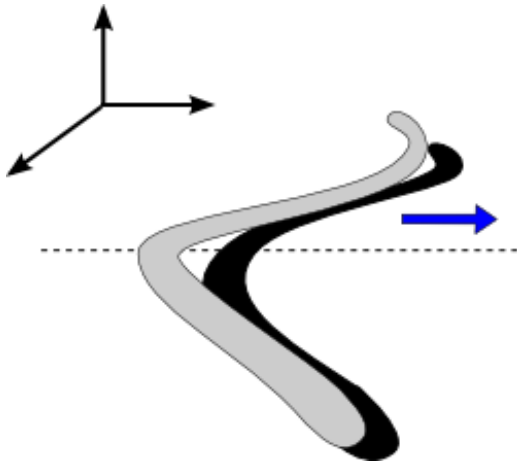
Locomotion gaits (III): Lateral shifting

Side-winding

- Characterization:

$$\alpha_v \rightarrow 0, k_v = k_h = k, \Delta \Phi_{vh} = 90$$

- DOF: α_h, k



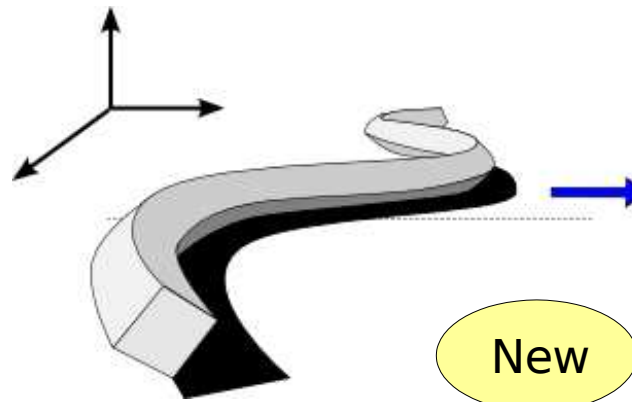
Inclined side-winding

- Characterization:

$$\alpha_h = \alpha \cos \beta, \alpha_v = \alpha \sin \beta$$

$$k_v = k_h = k, \Delta \Phi_{vh} \rightarrow 0$$

- DOF: α, β, k

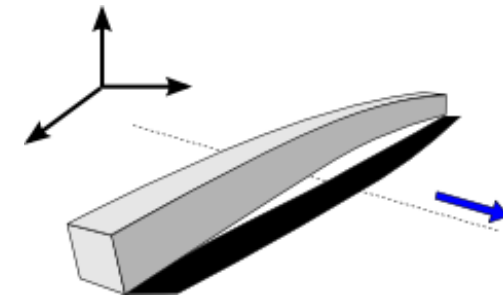


Flapping

- Characterization:

$$\alpha_h = \alpha_v = \alpha, \alpha > \alpha_{min}, \Delta \Phi = 90$$

- DOF: α



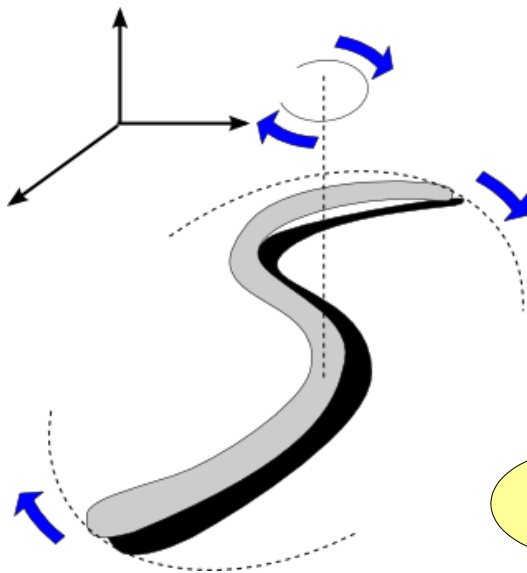
Locomotion gaits (IV): Rotating

S shaped rotation

- Characterization:

$$\alpha_v \rightarrow 0, k_v = 2k_h, \Delta\Phi_{vh} = 0$$

- DOF: α_h, k_h

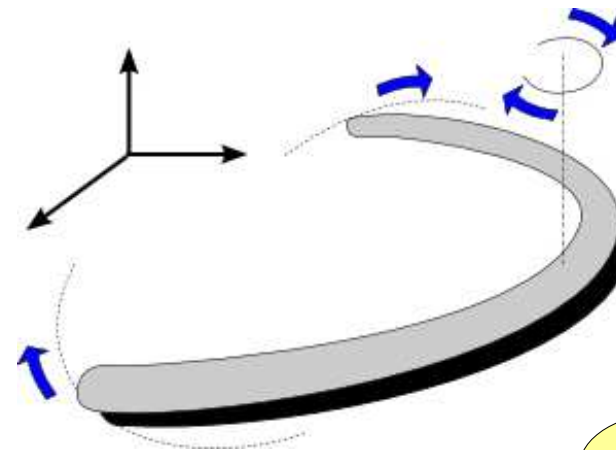


U-shaped rotation

- Characterization:

$$\alpha_v \rightarrow 0, \Delta\Phi_{vh} = 90$$

- DOF: α_h, k_v



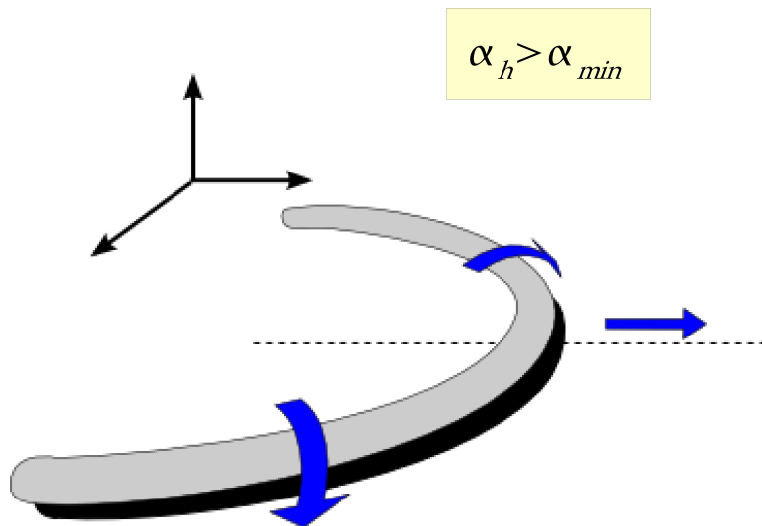
Locomotion gaits (V)

Rolling

- Characterization:

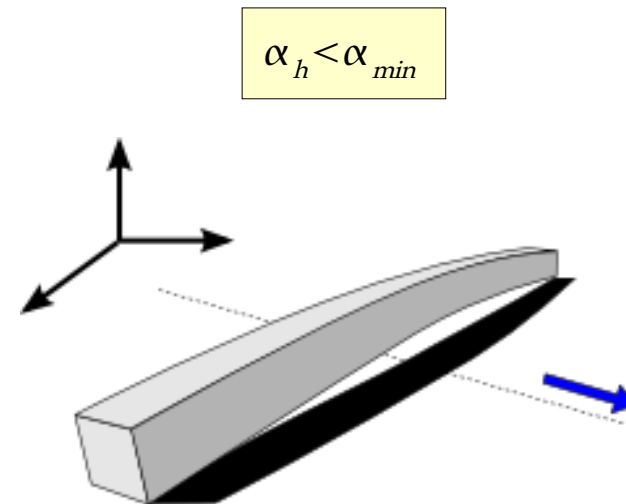
$$\alpha_v = \alpha_h, \alpha_h > \alpha_{min}, \Delta \Phi_{vh} = 90$$

- DOF: α_h



Flapping

- If the section is a square, the flapping gaits is achieved when:

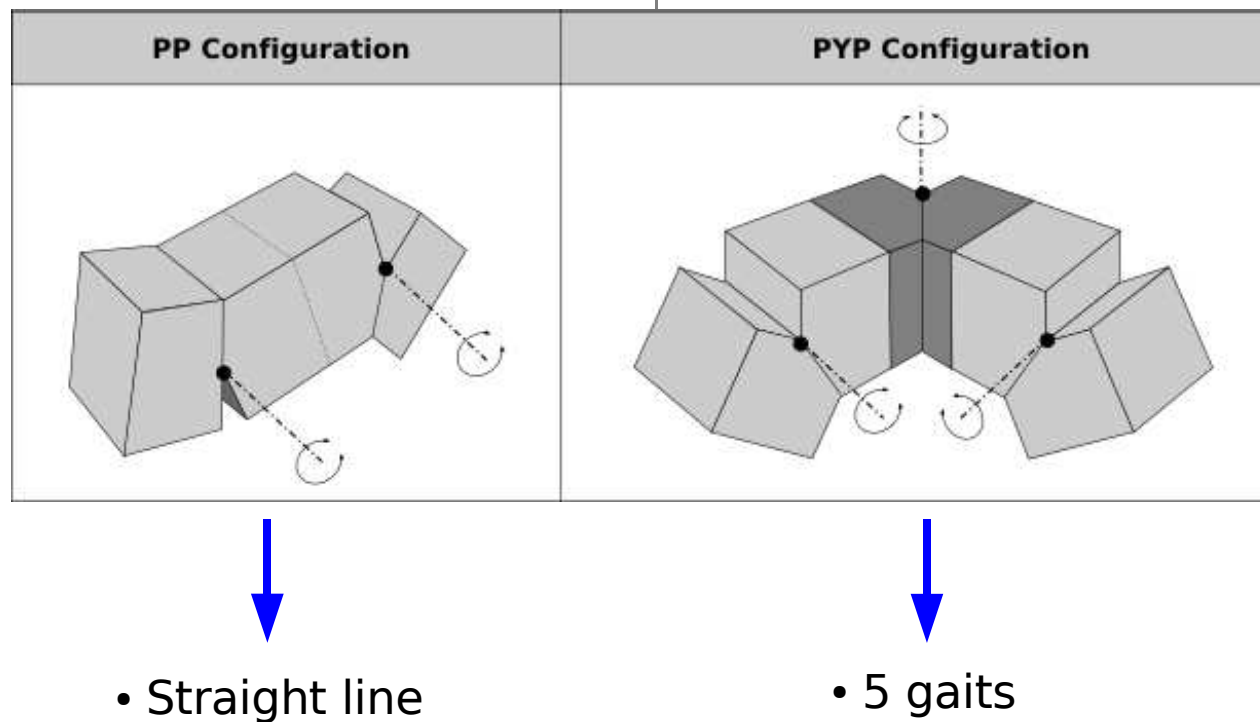


Outline

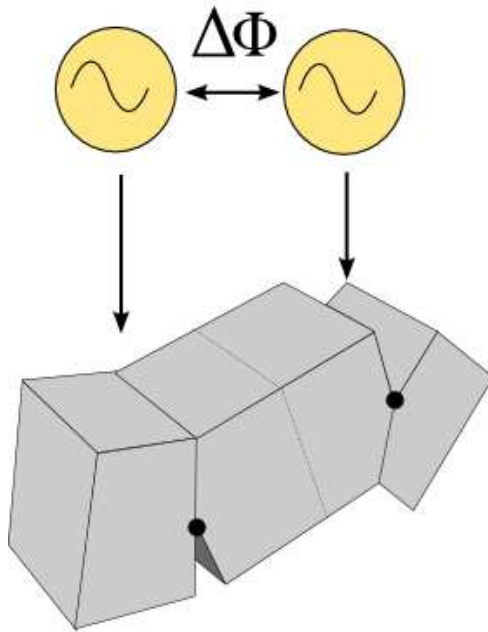
1. Introduction and objectives
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Minimal configurations

- Configurations with the minimal number of modules that are able to move
- Searching the control space using genetic algorithms

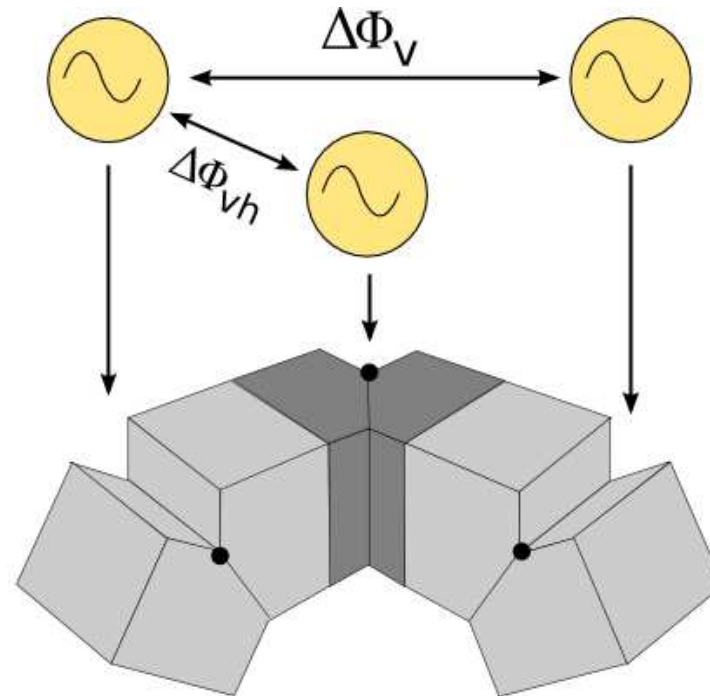


Control space



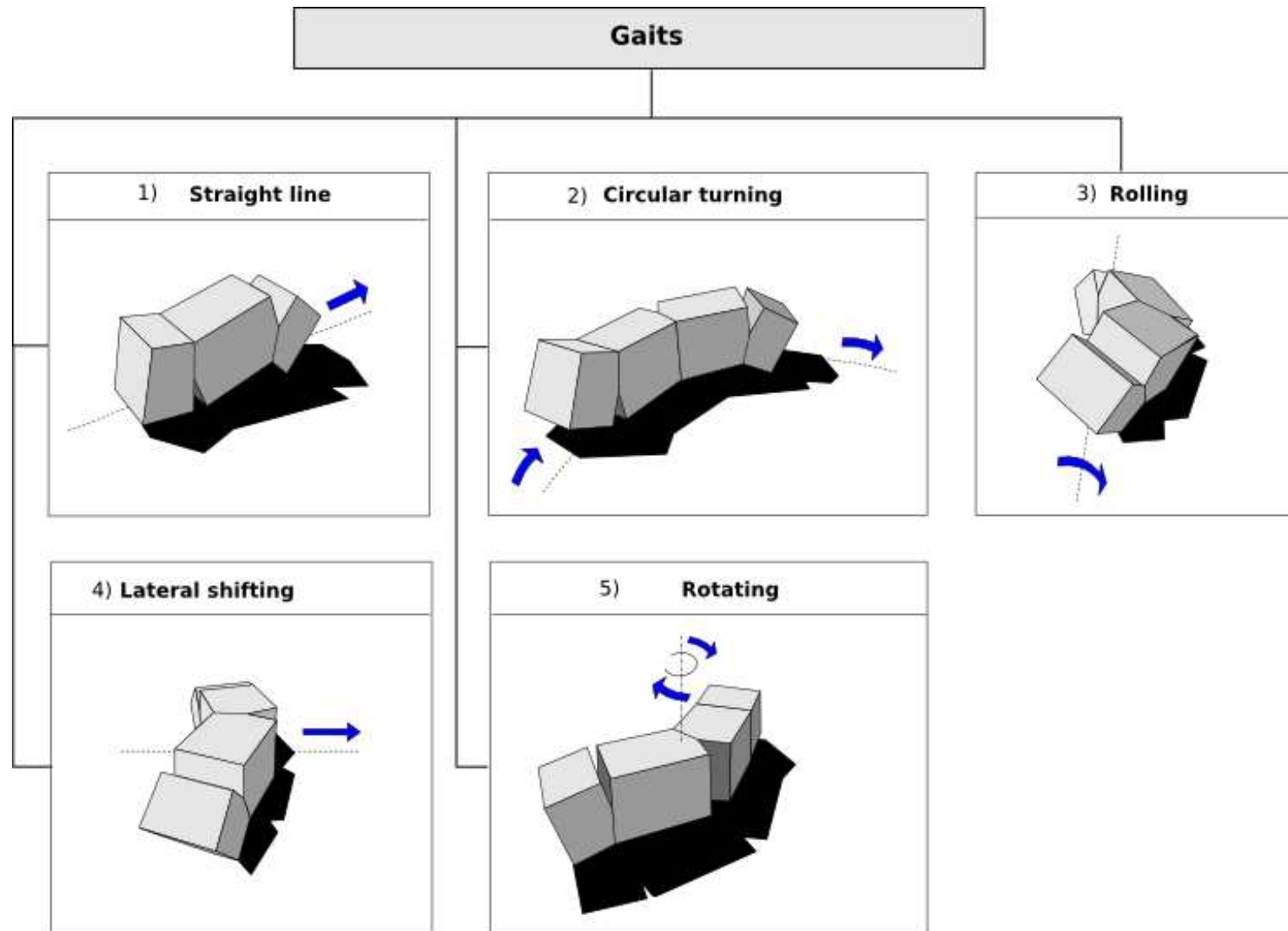
- 2 dimensions space

$$A, \Delta\Phi$$



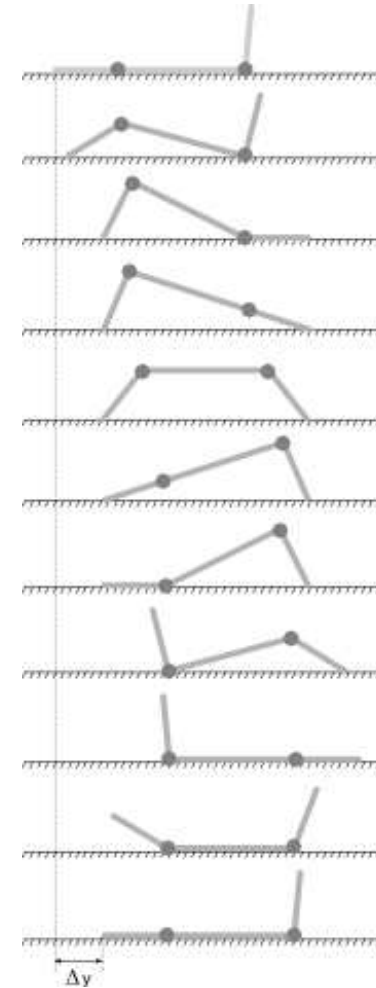
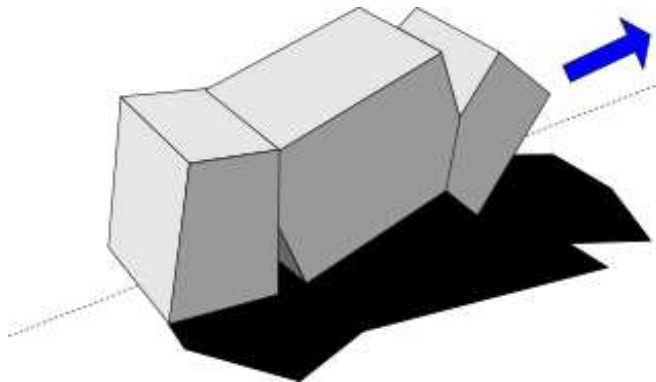
- 4 dimensions space

$$A_v, A_h, \Delta\Phi_v, \Delta\Phi_{vh}$$



Locomotion gaits (II): Straight line

- Study of the wired-model
- Best coordination: $\Delta\Phi \in [108, 110]$
- Maximum step: Best coordination + $A=90$



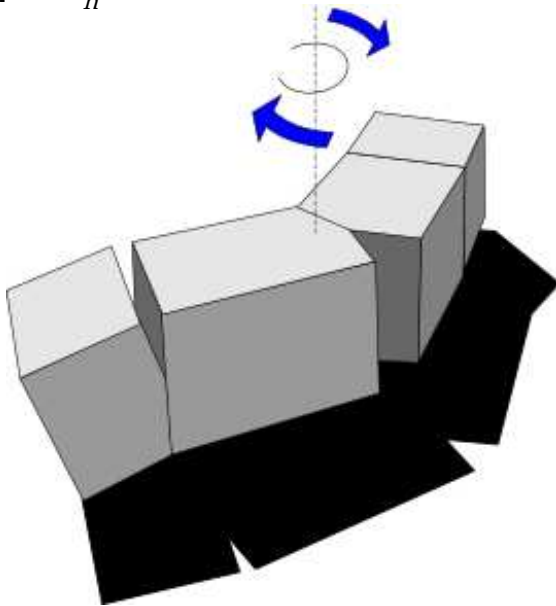
Locomotion gaits (III)

Rotating

- Characterization:

$$A_v \rightarrow 0, \Delta \Phi_v = 180, \Delta \Phi_{vh} = 90$$

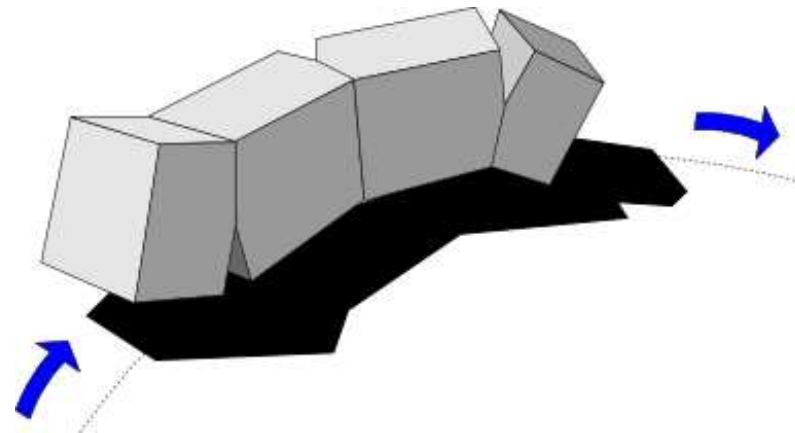
- DOF: A_h



Circular turning

- Characterization: $\varphi_h = A_h$

- DOF: $A_v, A_h, \Delta \Phi_v$



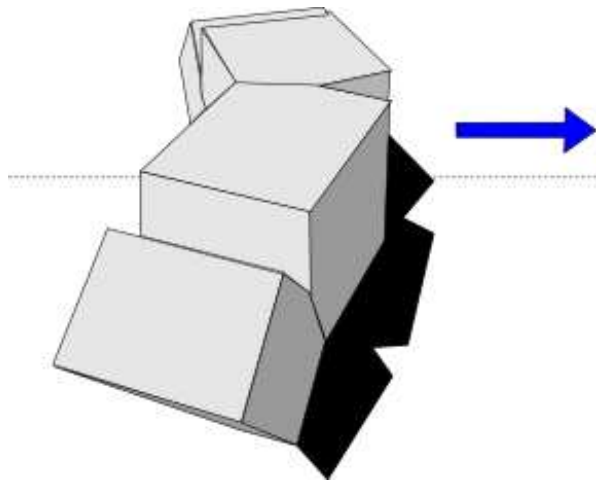
Locomotion gaits (VI)

Lateral shifting

- Characterization:

$$A_v \rightarrow 0, \Delta \Phi_v = 0, \Delta \Phi_{vh} = 90, A_h < A_{min}$$

- DOF: A_h

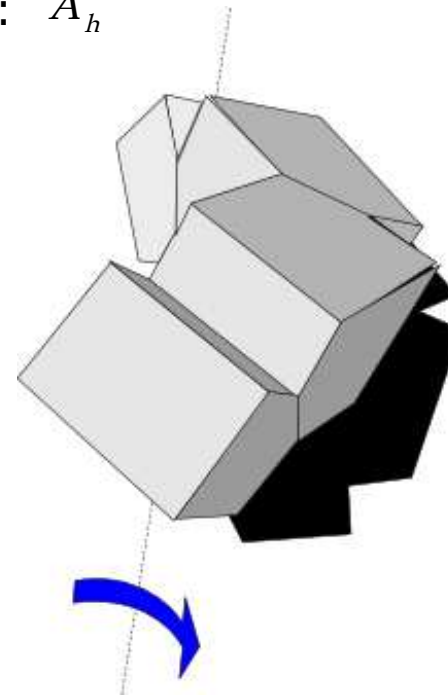


Rolling

- Characterization:

$$A_v = A_h, \Delta \Phi_v = 0, \Delta \Phi_{vh} = 90, A_h > A_{min}$$

- DOF: A_h

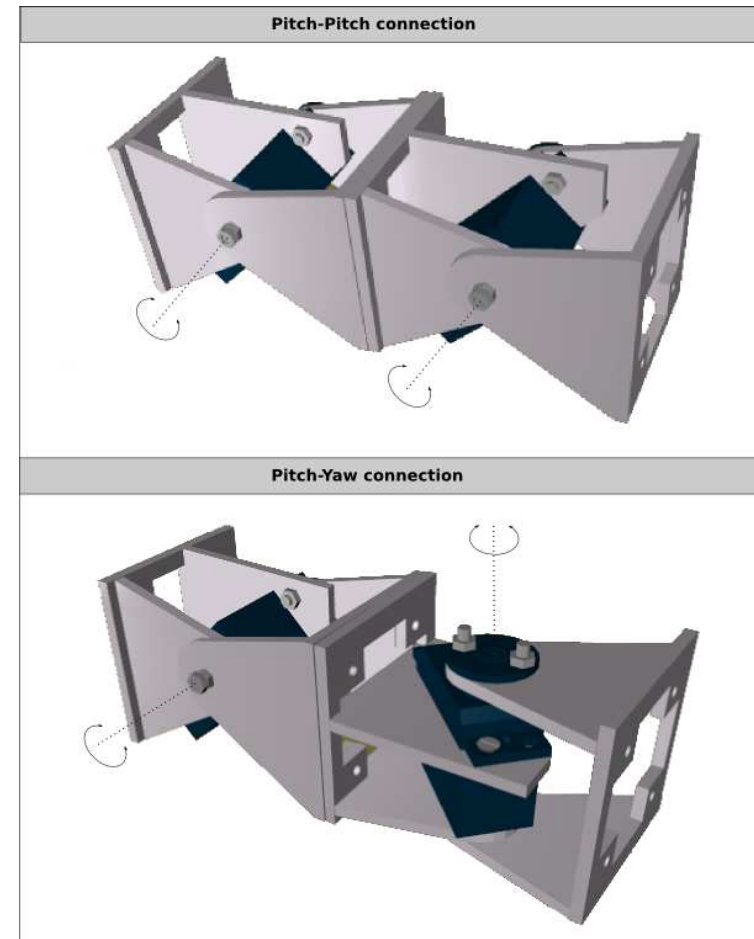
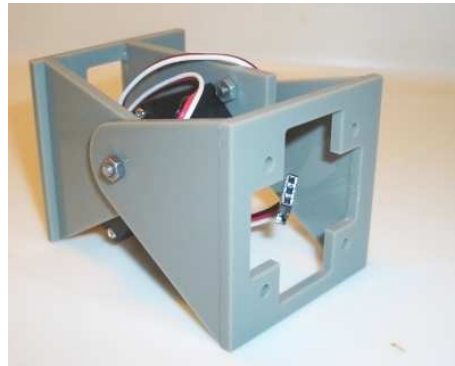
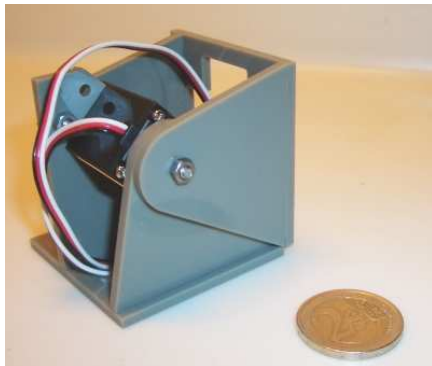


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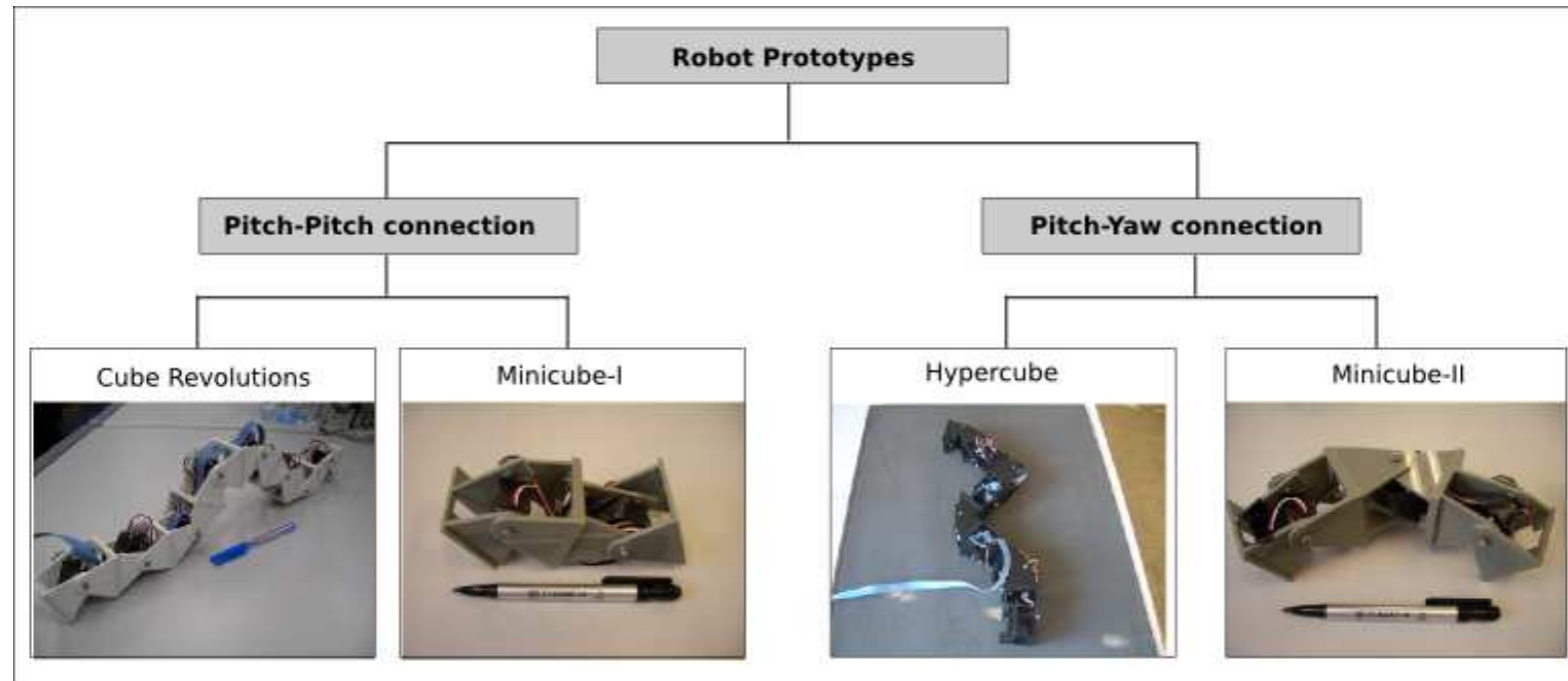
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Y1 Modules

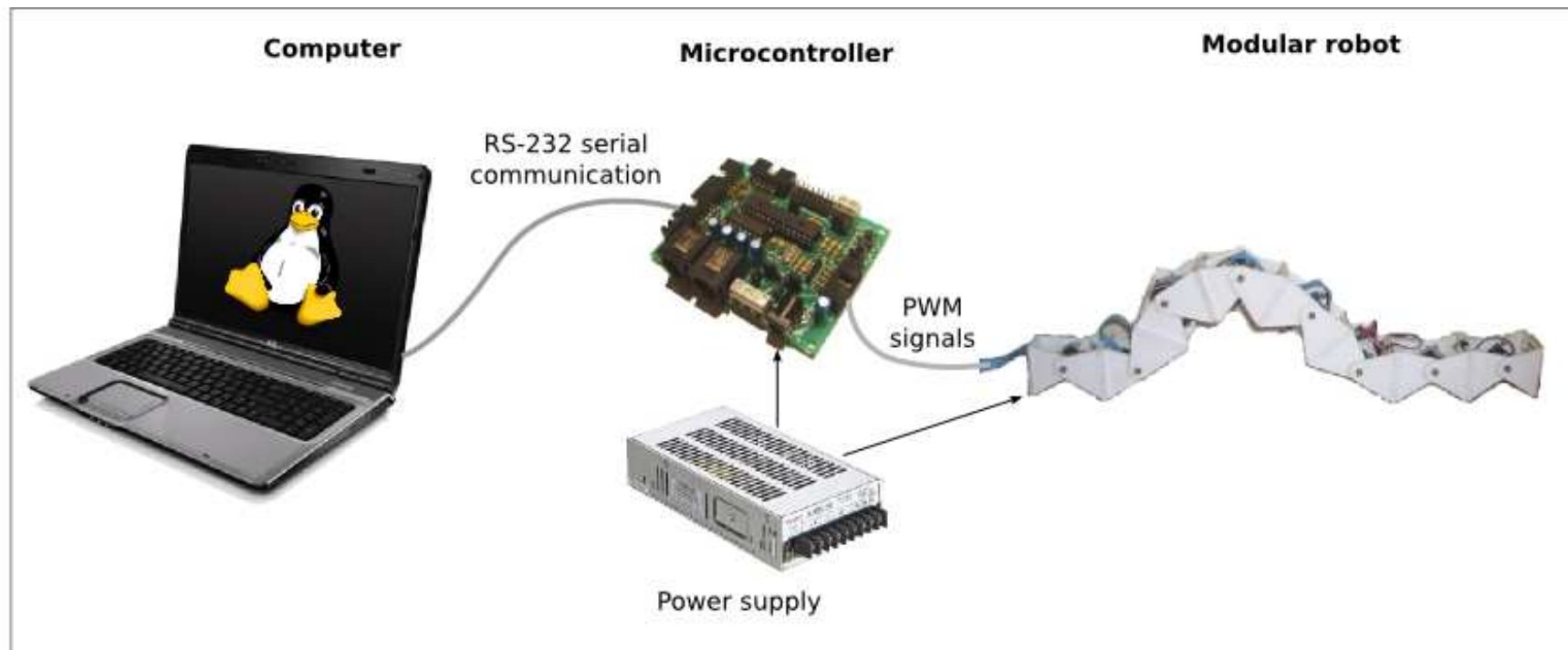
- One degree of freedom
- Easy to build
- Cheap
- Open and “Free”



Modular Robots Prototypes

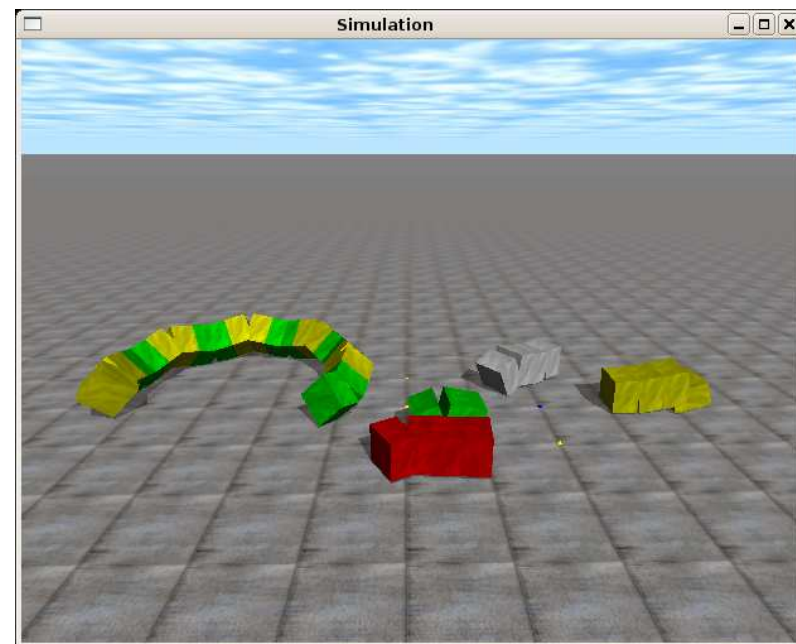
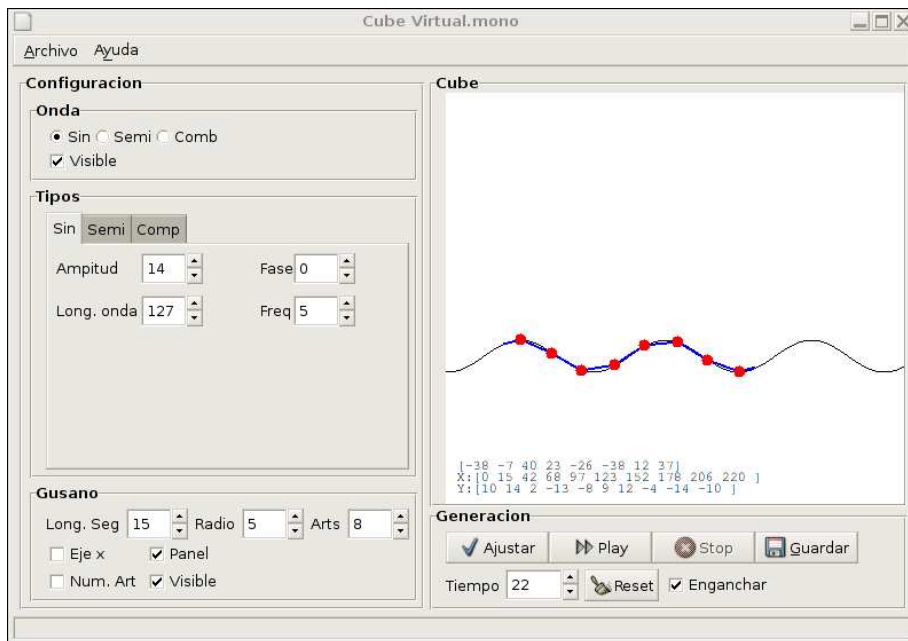


Robot controlling



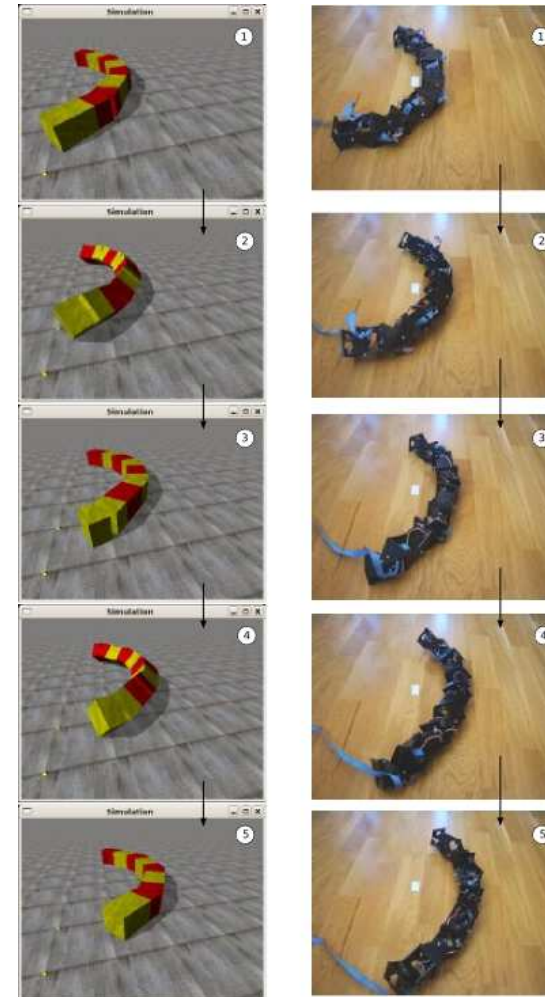
Software

- 1D topology simulator (Based on Open Dynamics Engine [ODE])
- Generics algorithms: PGAPack
- Mathematical models in Octave/Matlab



Experiments description

- **Simulation:**
 - Continuous model locomotion
 - Discrete model locomotion
 - Minimal configurations
 - Data collection
 - Comparison with the mathematical model
- **Real Robots:**
 - Locomotion validation
 - Bad locomotion solutions elimination



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Summary

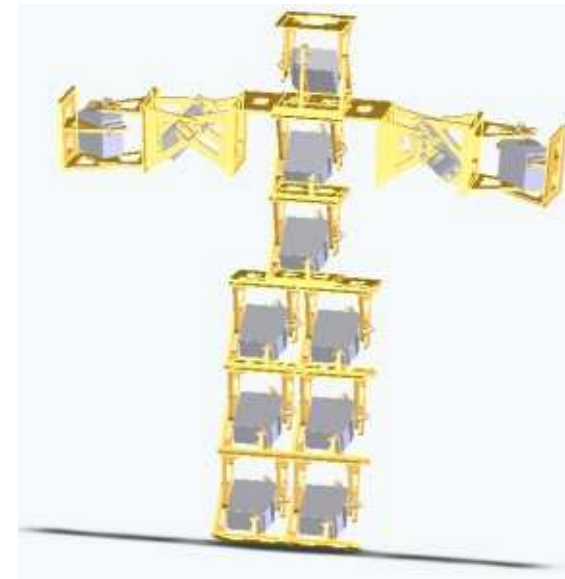
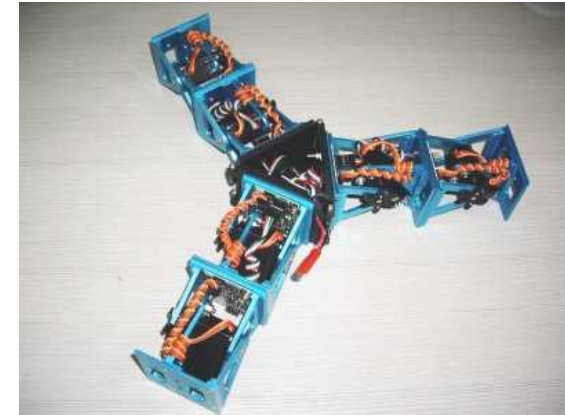
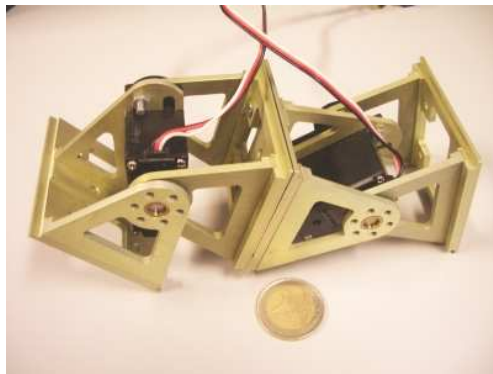
- State of the art reviewing
- Modular robot classification established
- Starting hypothesis: Sinusoidal generators
- Mathematical models development
- Solution searching (genetic algorithms)
- Development of a simulator
- Model comparison (experimental vs mathematical)
- Design of a robotic platform
- Experiments on real modular robots

Main contributions

- **Viability** of the Sinusoidal generator controlling model
- At least, **5 different gaits** can be achieved
- Control space **minimal dimensions**: 2 y 5
- **3 new locomotion gaits**: U and S shaped rotation and inclined side-winding
- **Minimal configurations**
- **Relationships** between the generators and the kinematics
- A **new open modular robotic platform** (modules, hardware and software)
- Knowledge summarized into **27 Key Locomotion principles**

Future work

- Dynamics and Energetic models
- Sensors feedback
- New modules: **GZ-I**
- 2D Topologies locomotion
- Application to climbing caterpillars
- Behaviours implementation (JDE)



Publications

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Modular Robotics and Locomotion: Application to Limbless Robots

Juan González Gómez
Ph.D. Thesis



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